

Short summary: Develop a fusion-based approach to reliably detect small, low-RCS objects at sea by combining radar and camera data, and use radar cues to streamline image labeling.

Summary

Conventional marine radars alone struggle to detect small, low radar-cross-section (RCS) objects (like kayaks, swimmers, buoys, pylons) in higher sea states and heavy sea clutter, and optical camera detectors are limited by scarce labeled data in marine settings. By fusing X-band radar and cameras, the system will improve the detection of small maritime objects and automatically assist in labeling camera images using radar returns as cues.

SEA-ING IS BELIEVING: AUTOMATED MARINE OBJECT LABELING

The first look.

PART 1. INTRODUCTION & BACKGROUND

In maritime settings, vision-based object detection faces a severe data labeling bottleneck. Building large, annotated image datasets of small boats or floating objects at sea is expensive and time-consuming. Such objects, as seen in Figure 1, are relatively rare and often only a few pixels in size, so collecting and labeling examples requires expert annotators to sift through hours of mostly-empty footage.



Figure 1: Small Object Detection: A Succinct Review

Existing public maritime image datasets are few, and those that do exist have limited scale and diversity. Deep learning models demand plentiful labeled examples to achieve high accuracy, yet acquiring those examples on the open ocean is impractical. As a result, current vision models trained on sparse maritime data struggle to detect small, distant objects reliably. In comparison to common object detection on land, labeling maritime scenes (mostly water with tiny far-off targets) is far more costly and difficult.

This shortage of annotated data is a major barrier to deploying robust camera-based detectors on marine platforms.

Secondary sensor suites, such as marine radar systems, often have difficulty reliably detecting very small, low-RCS targets (e.g. a partially submerged person, small buoy, or debris) against the dynamic ocean background. Such objects return very weak echoes that are easily masked by sea clutter, the radar returns from waves. In practice, the unfavorable signal-to-clutter ratio means that small craft or persons in the water may go undetected by conventional X-band navigation radars in rough seas. Even specialized surveillance radars (for example, airborne periscope-detection radars) suffer high false-alarm rates in choppy conditions, as breaking wave tops produce spurious bright clutter spikes that can mimic or eclipse genuine targets. In short, using radar alone, small maritime objects are often lost in clutter or generate many false detections. An example of Low-RCS objects in differing clutter environments is provided in Figure 2.

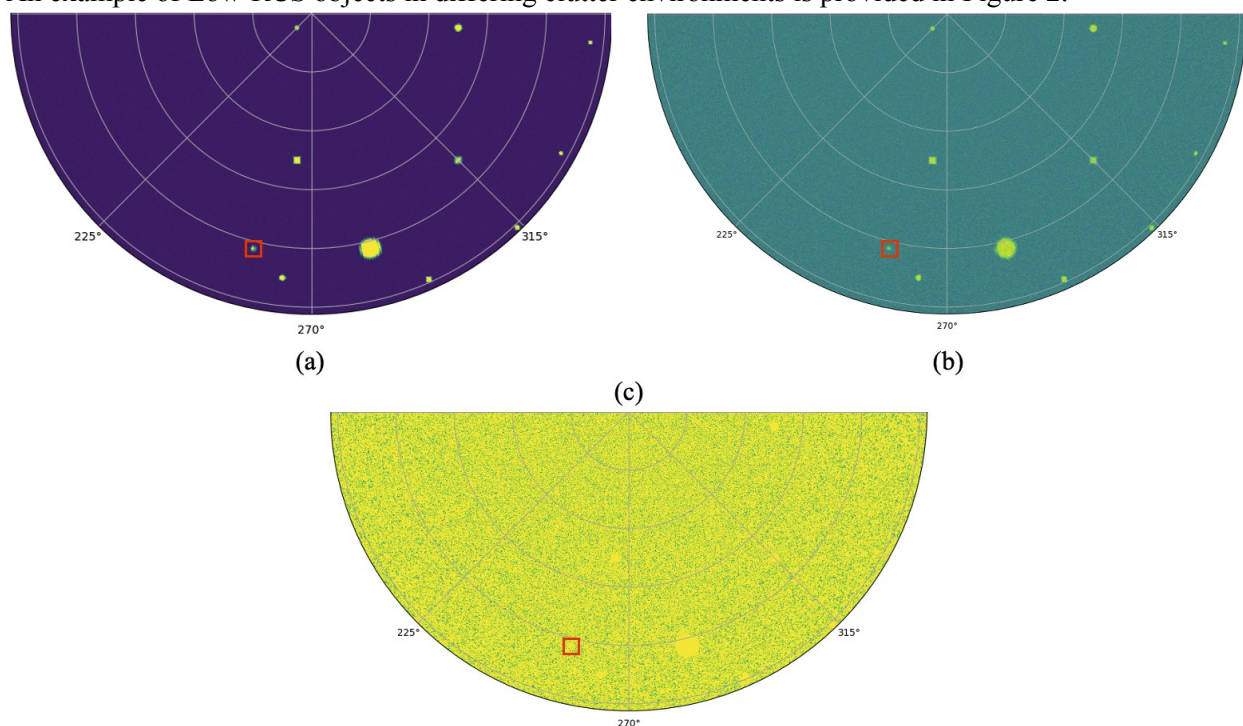


Figure 2: Low-RCS Objects in Low, Med, & High Rain Clutter

This represents a critical gap for maritime safety and security, as low-profile hazards or intruders may elude single-sensor detection. However, Radar-camera sensor fusion offers a promising solution to overcome these challenges. Marine radar operates day or night and in fog, rain, or glare where optical cameras often fail, and can detect objects at long range beyond the visible horizon. A preliminary sketch of this proposal is seen in Figure 3.

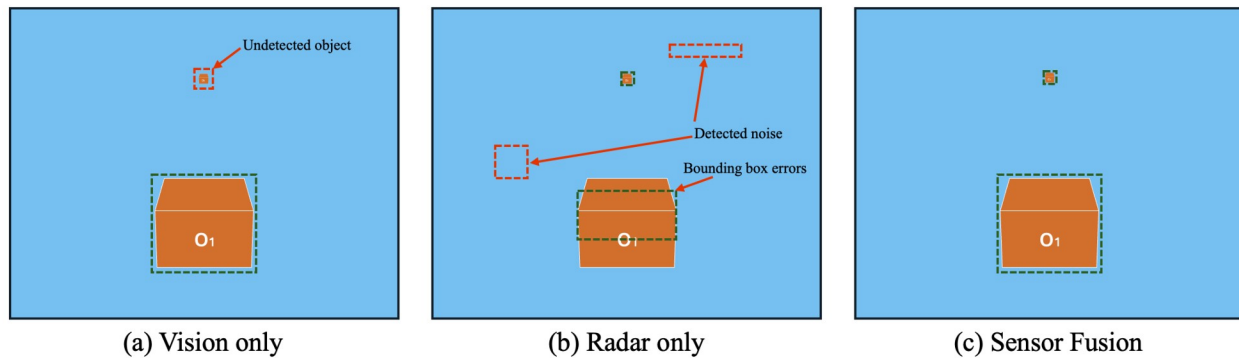


Figure 3: Comparison of Camera and Radar detection systems

By leveraging radar’s all-weather detections to cue the camera, we can effectively “seed” the image labeling process. For instance, if the radar returns indicate a target at a certain range and bearing, that can flag a region-of-interest in the corresponding camera frame for a human annotator (or an assisting algorithm) to verify and label. Researchers have begun exploring such cross-sensor cueing; for example, a Radar Region Proposal Network (RRPN) has been proposed to project radar detections into the camera view as candidate object regions. In our context, the radar will act as an automatic proposal generator for the camera, focusing attention on likely object locations, especially valuable when objects are small or visibility is poor. The overarching goal (and motivation for this project) is to bridge the radar and optical modalities to overcome their individual limitations. A fusion of radar and camera information promises more robust detection of small maritime objects and a means to automate data labeling. Radar can compensate for the camera’s shortcomings in low-visibility conditions, while the camera provides high-resolution classification that radar lacks. By jointly using radar and optical sensors, we aim to improve the detection performance for small objects at sea and develop an automated labeling aid; addressing both the detection gap of radar and the data scarcity problem of deep learning models. One possible annotator-assisted pipeline is shown in Figure 4; combining 2 separate algorithms to create useful annotation data.

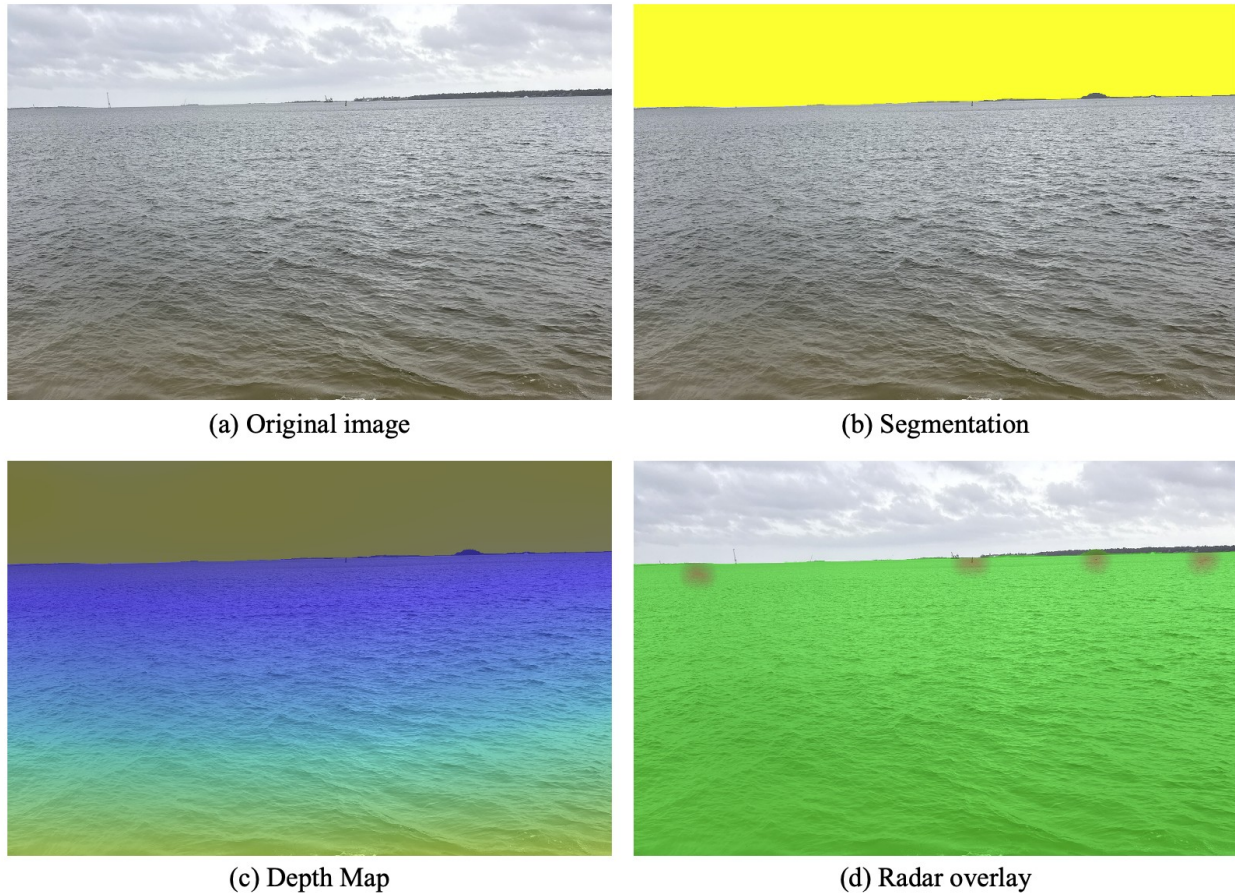


Figure 4: Radar-Overlay Object Detection Pipeline

PART 2. TECHNICAL BACKGROUND

Part 2.1 Marine Radar Fundamentals

Modern marine radars operate using either a pulsed or Frequency-Modulated Continuous-Wave (FMCW) scheme. Traditional pulsed radar (often magnetron-based) transmits high-power microwave pulses and measures the echo return time to determine target range. These can emit kilowatt-level peak power pulses, offering long-range detection, but they require a warm-up period and have a short blind zone (minimum range) during each pulse when the radar cannot receive. In contrast, FMCW radar continuously sweeps its transmit frequency and listens for the frequency-shifted returns, achieving similar range performance with much lower peak power (on the order of tens of watts). An example of the difference between FMCW and Pulse radar is shown in Figure 5. FMCW radars have no inherent blind range (since they transmit and receive simultaneously) and provide improved short-range detection and Doppler sensitivity due to their coherent processing. While magnetron pulse radars remain popular for their simplicity and long-range capability, solid-state FMCW radars are increasingly adopted for their instant-on operation, lower power consumption, and ability to detect very close-in objects that pulsed radars might miss.

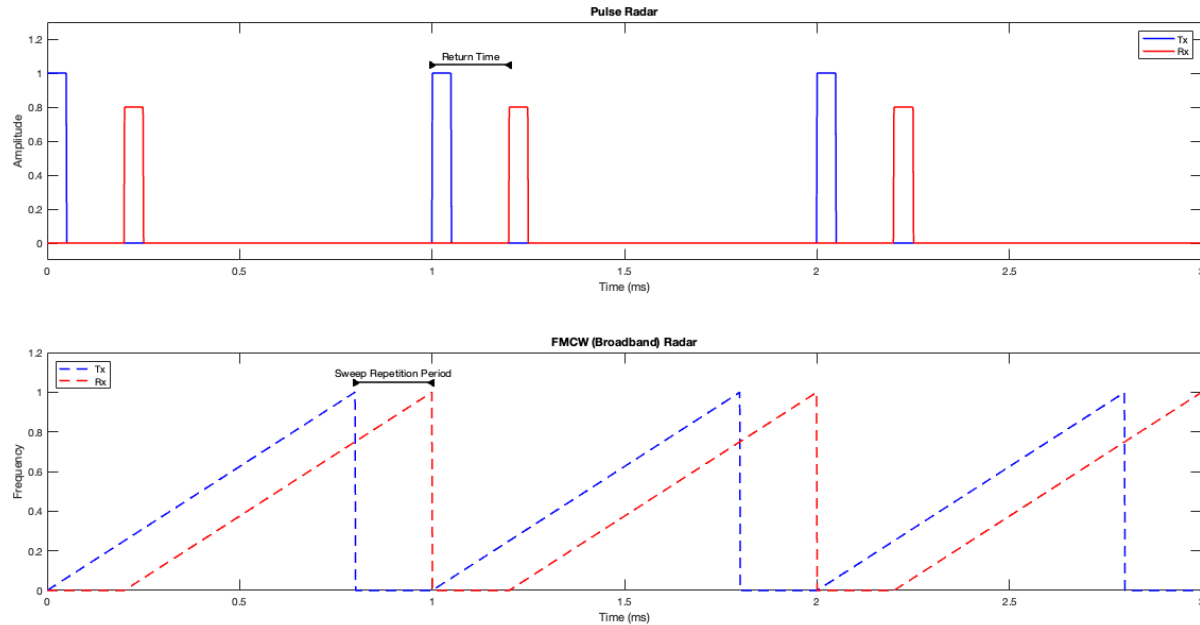


Figure 5: Comparison of Pulse and FMCW Radar Waveforms

For this project, an X-band marine radar (9.4 GHz) is utilized due to low-cost, power efficiency, and ease-of-use. Regardless of the chosen architecture, understanding radar operating mode (pulsed vs. FMCW) will inform how collected data is to be processed and integrated with other sensors or cameras. Although some radar systems (automotive primarily) are stationary; the majority of marine radar provides a 360° panoramic image of echoes (often referred to as PPI or Plan Position Indicator) in which targets are detected against sea clutter.

Part 2.2 Gain Control and CFAR

A key challenge in radar processing is distinguishing small targets from sea clutter and noise, which requires careful setting of detection thresholds. Radar units include gain control circuits to adjust the amplification of incoming echoes, effectively controlling the display intensity of returns. Proper gain setting is crucial: set the gain too low and weak targets disappear; set it too high and the display saturates with background noise or sea clutter. On older or lower-end radars, the gain is adjusted manually by the operator; typically increasing it until noise speckles just become visible, then dialing back slightly. Skilled operators constantly tweak the gain to maximize target visibility without overwhelming clutter. In heavy clutter (e.g. high sea state or rain), an operator may reduce gain to avoid false returns, at the cost of possibly losing very small echoes. A simple example of high and medium clutter environments is shown in Figure 6. As can be seen in Figure 6, although a gain setting can work in one environment, it does not necessarily guarantee it will work in another environment. Thus the major drawback of manual tuning is its subjectivity and inclination towards continuous readjustment.

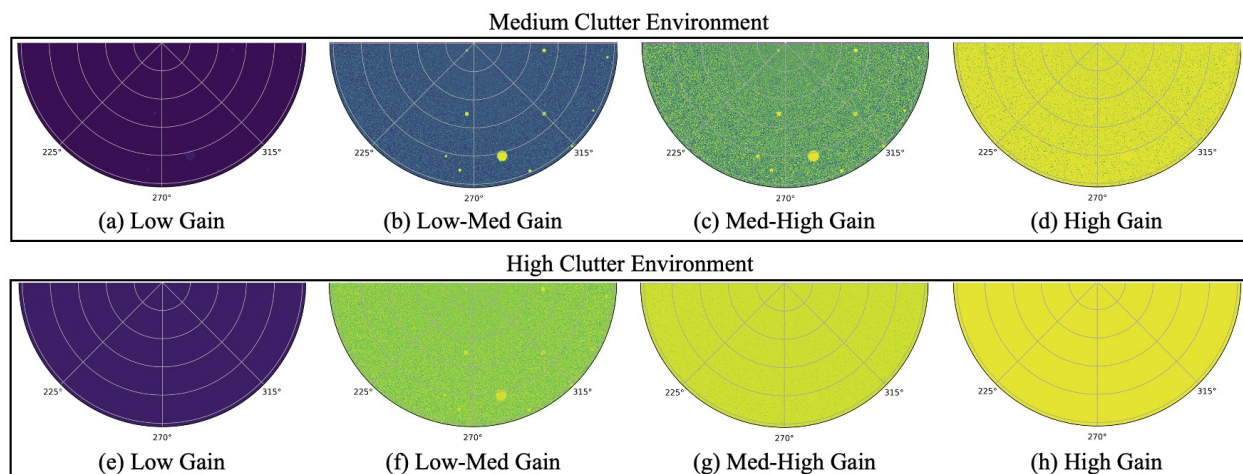


Figure 6: High and Medium Clutter Environments with Varying Gain Levels

Modern radar systems (such as the commercial ones utilized for this project) often employ Automatic Gain Control (AGC) to continually adapt receiver gain based on overall return levels. AGC will lower gain when the radar sees a lot of strong returns (for instance, large waves or nearby land) to prevent the display from washing out, and raise gain in calmer conditions to maintain sensitivity. This adaptive approach keeps the output within an optimal range without operator intervention. However, AGC is a blunt instrument; it does not discriminate between wanted targets and unwanted clutter. In the presence of heavy clutter, AGC may inadvertently suppress small target echoes along with the background, since it simply equalizes the total output level. Thus, a tiny kayak could be damped out when large waves are in the vicinity, as the radar autonomously reduces gain to handle the clutter. Finding the right balance is difficult, which is one reason we are motivated to handle raw radar data in a more intelligent way (using sensor fusion and learning, as described later) rather than relying purely on built-in AGC.

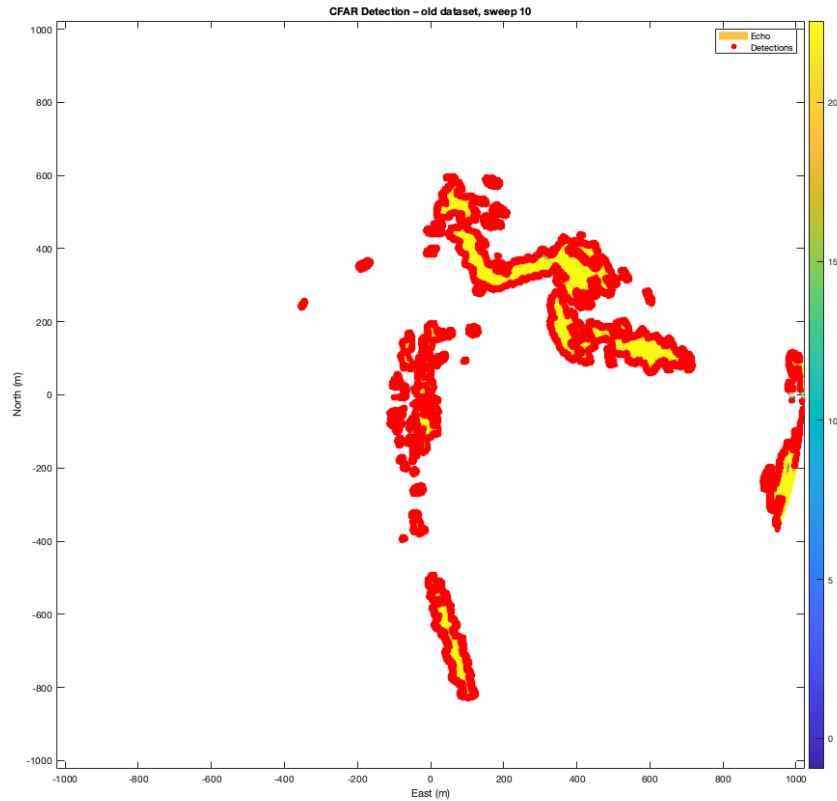


Figure 7: CFAR Implementation for Data Collected at Lake Murray

Another fundamental radar processing technique is the Constant False Alarm Rate (CFAR) detector. The elementary basics of CFAR is shown graphically in Figure 8 while a version of CA-CFAR is shown on real data in Figure 7. CFAR adaptively sets the detection threshold based on local clutter statistics to maintain a roughly constant false alarm probability. In practice, the radar computes the average or distribution of the return power in a sliding window around a cell and chooses a threshold such that, say, only one in a million noise cells exceeds it (for a false alarm rate of 10^{-6}). Any echo stronger than this threshold is declared a target. CFAR is very useful in maritime radar since sea clutter is not uniform: it raises thresholds in regions of heavy backscatter (e.g. breaking waves) and lowers thresholds in calmer areas. This adaptivity prevents a flood of false alarms that a fixed threshold would trigger in rough seas, and conversely, it retains sensitivity in calm waters where a fixed high threshold would miss targets.

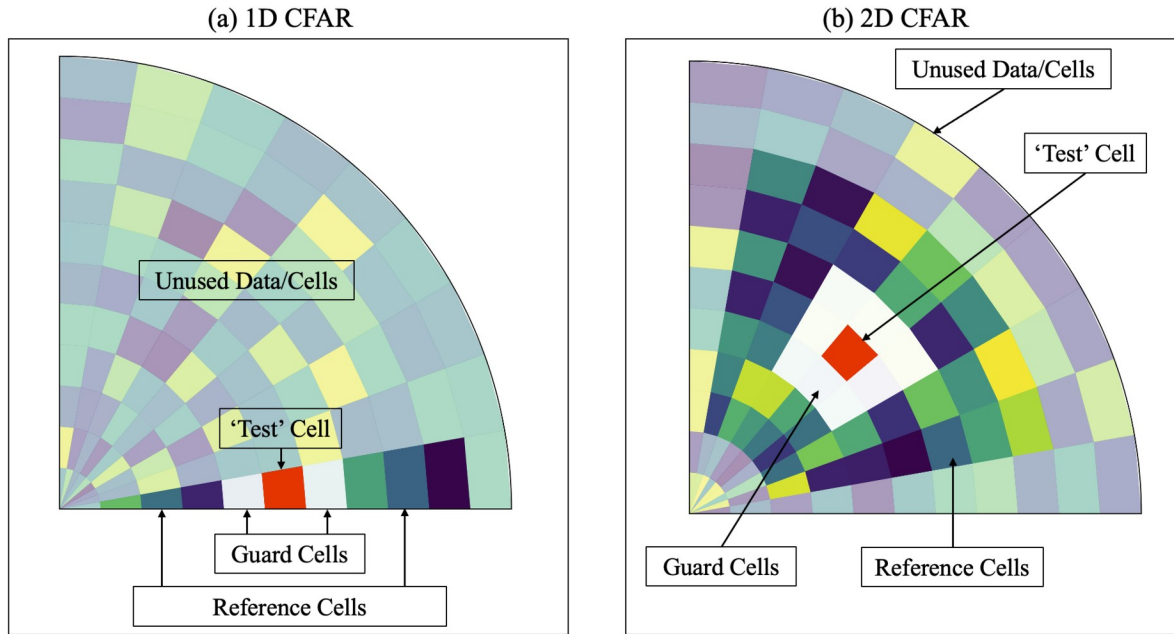


Figure 8: CFAR 1D and 2D on Sample Radar Data

In addition to requiring tuning for each radar, CFAR is not foolproof. In intense, non-homogeneous clutter, a small target's echo may be submerged below a heightened local threshold; the radar effectively raises the bar so high (to keep false alarms down) that the target is missed. For example, a slow-moving person in water surrounded by choppy waves might be invisible to a standard CFAR detector because the local sea clutter drives the threshold above the person's return level. On the flip side, in non-homogeneous clutter scenarios, CFAR can become overly sensitive, requiring careful tuning of the false-alarm rate to avoid peppering the screen with false targets. These limitations motivate exploring machine learning approaches, which can potentially learn to recognize the subtle signal of small objects within clutter and motivate using additional sensor modalities (camera) to confirm which radar returns are real targets.

Part 2.3 Planned Sensor Fusion Approach

By combining radar and camera data, it is believed there is a path to capitalize on the strengths of each. The radar provides range, bearing, and a detection confidence (echo amplitude) for objects, even in darkness or haze. The camera provides rich appearance information to classify and localize those objects in image space. A critical step in fusion is sensor co-registration; i.e. calibrating the coordinate frames of the radar and camera so that we can map a radar detection (range θ , bearing ϕ) to the corresponding pixel location in the camera image. A calibration technique has not been settled upon yet, but will likely be of the autonomous variety.

Once this calibration is in place, each radar detection can be projected into the camera view as an annotated bounding box or marker. This allows a few immediate benefits: (1) Improved detection - if the radar detects something that the camera-based algorithm missed, the system can direct attention to that region in the image (for a potential false-negative recovery); and (2) Assisted labeling - as we begin building a training dataset for camera object detection, the radar can pre-suggest regions of interest, drastically reducing the human effort needed to scan each frame for tiny objects. In essence, the radar acts as an "extra pair of eyes" for the camera, especially in scenarios of poor visibility or tiny targets. The aim is to implement a radar-assisted labeling pipeline where radar detections provide initial bounding boxes for objects in the

camera images. A human annotator (or an automatic verification step) would then confirm or adjust these boxes, resulting in labeled images with significantly less manual labor. A simple diagram with real sensor data overlaid by a possible implementation for human annotators is shown in Figure 9. This approach addresses the aforementioned labeling bottleneck by focusing human attention on likely object locations instead of erratically and aimlessly searching large empty frames for small objects.

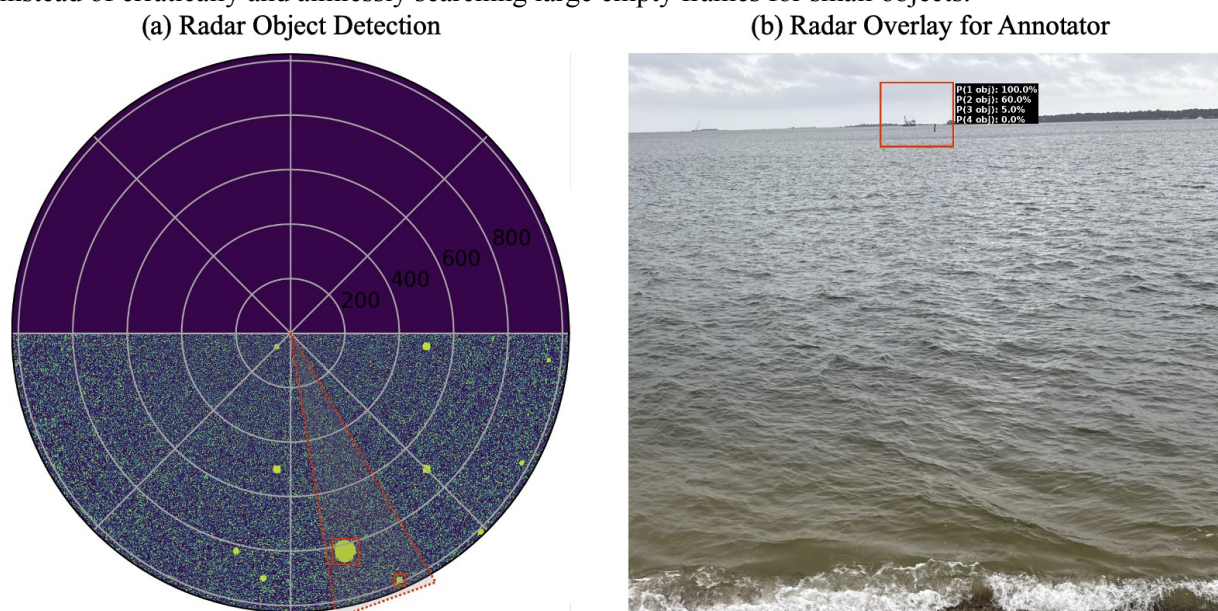


Figure 9: Annotator-Radar Labelling Pipeline

Part 2.4 Multi-Gain Radar Data for CNN

A novel aspect of the approach is how Radar gain is handled. As discussed, a single fixed gain (or even AGC) can either wash out weak targets or flood the display with clutter. Our solution posits to capture multiple radar intensity frames at different gain levels and feed them together into a neural network, allowing the network to learn a composite understanding of the scene. The marine radar we use can be configured to produce such multi-gain outputs via low-level manipulation of the radar software, and utilizing temporal-based techniques. By doing so, the chosen detection architecture can avoid committing to a single threshold early on. If the gain is too low, one frame will show almost nothing – even strong returns like land might barely appear. Conversely, if the gain is too high, another frame will show everything lit up (strong targets and noise speckle everywhere) so that true targets are hard to distinguish. In practice today, an operator must manually adjust gain to find a Goldilocks setting for each situation. To reduce this dependence on an expert user, we propose an automatic multi-gain fusion. Because the radar display’s native output tends to be binary (see Figure 10) at any given gain, effectively an aggressive thresholding that yields mostly 0 or 1 in the pixel matrix, no single setting can capture all targets. By combining multiple “exposures” of the radar at low, medium, and high gain, it is possible to obtain a richer representation of the underlying signal strength.

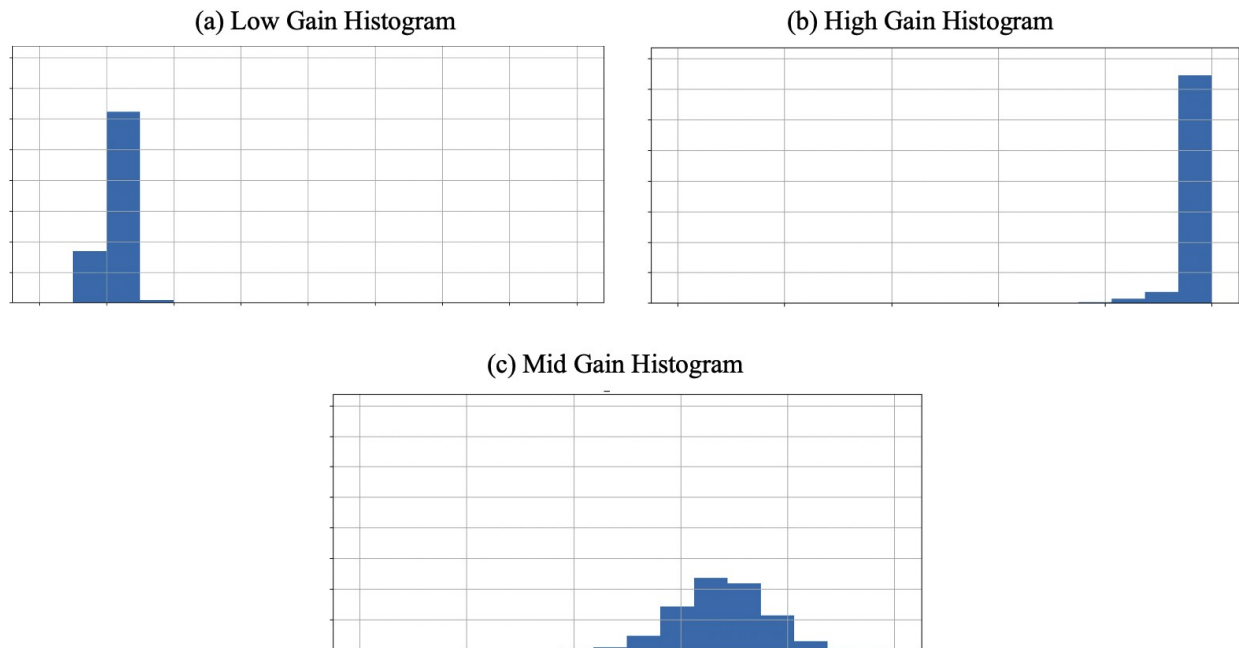


Figure 10: Histograms of Different Gain Settings – 4-bit Quantized

The neural network will receive a stack of these frames (e.g. as separate input channels), from which it can deduce that an object which is invisible in the low-gain frame but pops out in the high-gain frame is likely a small target in heavy clutter, whereas something visible in low gain but blurred in high gain might be a strong target or interference. Feeding multiple gain levels into a CNN is expected to improve small object detection, essentially providing the network with dynamic range context that the radar's own display thresholding throws away. This idea is inspired by the concept of multi-exposure imaging in photography, where combining images taken at different exposures can reveal details in both dark and bright regions. In our case, the “exposures” are radar gains. An initial convolutional model (e.g. a custom ConvNet or an adaptation of YOLO for radar) will be developed to take these multi-gain radar frames and output detections (range, angle) of potential targets. Those detections can then be mapped onto the camera images as described. Ultimately, the output of the fusion system will be both a radar PPI with target markers and camera frames with annotated objects; a comprehensive situational picture. Preliminary results for gain-combining and clutter removal under High clutter simulated settings is shown in Figure 11 with somewhat promising results.

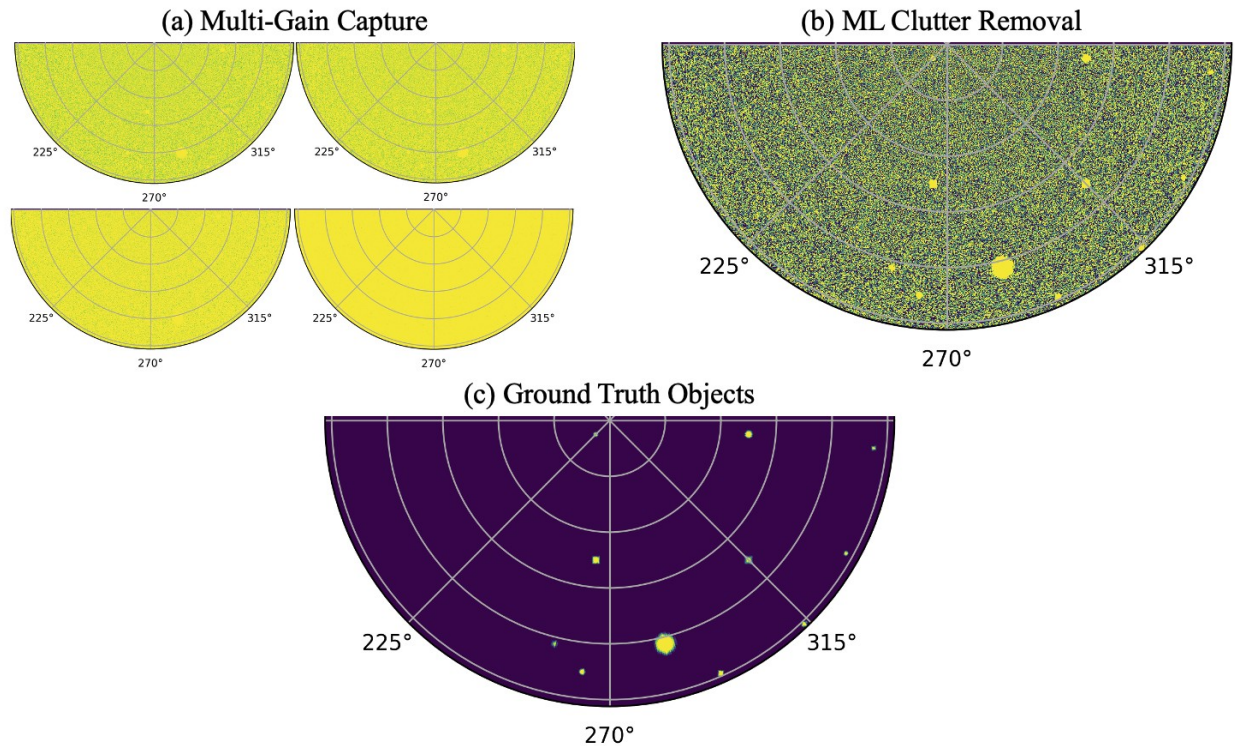


Figure 11: Multi-Gain Radar Detection for Stationary Low-RCS Objects in High Clutter Environments

PART 3. PLANNED WORK

Over the next several months, the plan is to design and implement an end-to-end radar-camera object detection pipeline that demonstrates the advantages of fusion and multi-gain radar processing. The major components and milestones of this research ‘phase’ are as follow below.

Part 3.1 Build the Sensor Rig

Set up the experimental platform combining a marine radar and an optical camera (and optionally an AIS receiver) on a test vessel or shoreline station. This includes hardware integration and calibrating the relative orientation and position of the radar and camera (i.e. performing the co-registration so that we can accurately overlay radar detections on camera images). Timing synchronization between sensors will be completed to ensure each radar sweep can be paired with the corresponding video frame. The current testing rig and a sketch of a proposed new testing rig is shown in Figure 12.

(a) Current Collection System



(b) Proposed EZ-Setup System



Figure 12: Radar Data Collection System - Current & Planned

Part 3.2 Collect Pilot Dataset

Conduct data collection trials to gather a pilot dataset of synchronized radar and camera recordings. The aim is to collect on less than 10 hours of radar PPI video (scans) with corresponding camera footage in various conditions. These will include scenarios with known small targets (e.g. a kayak, buoy, pylons) as well as background-only runs. Importantly, we will record raw radar data without lossy compression. The pilot dataset will be curated as A-MARV-pilot, a preliminary version of what will become a larger Amplitude-only Maritime Radar Video (A-MARV) dataset. All data will undergo minimal preprocessing: filtering obvious noise, applying our multi-gain transformation on each radar frame, and storing aligned radar-camera pairs for training and analysis. The initial data is planned to be collected at Lake Murray with future locations to include SC lakes and shorelines before moving to more complex and diverse sea scenarios. The full selection of lakes planned for the initial release of lake data is shown in Figure 13.

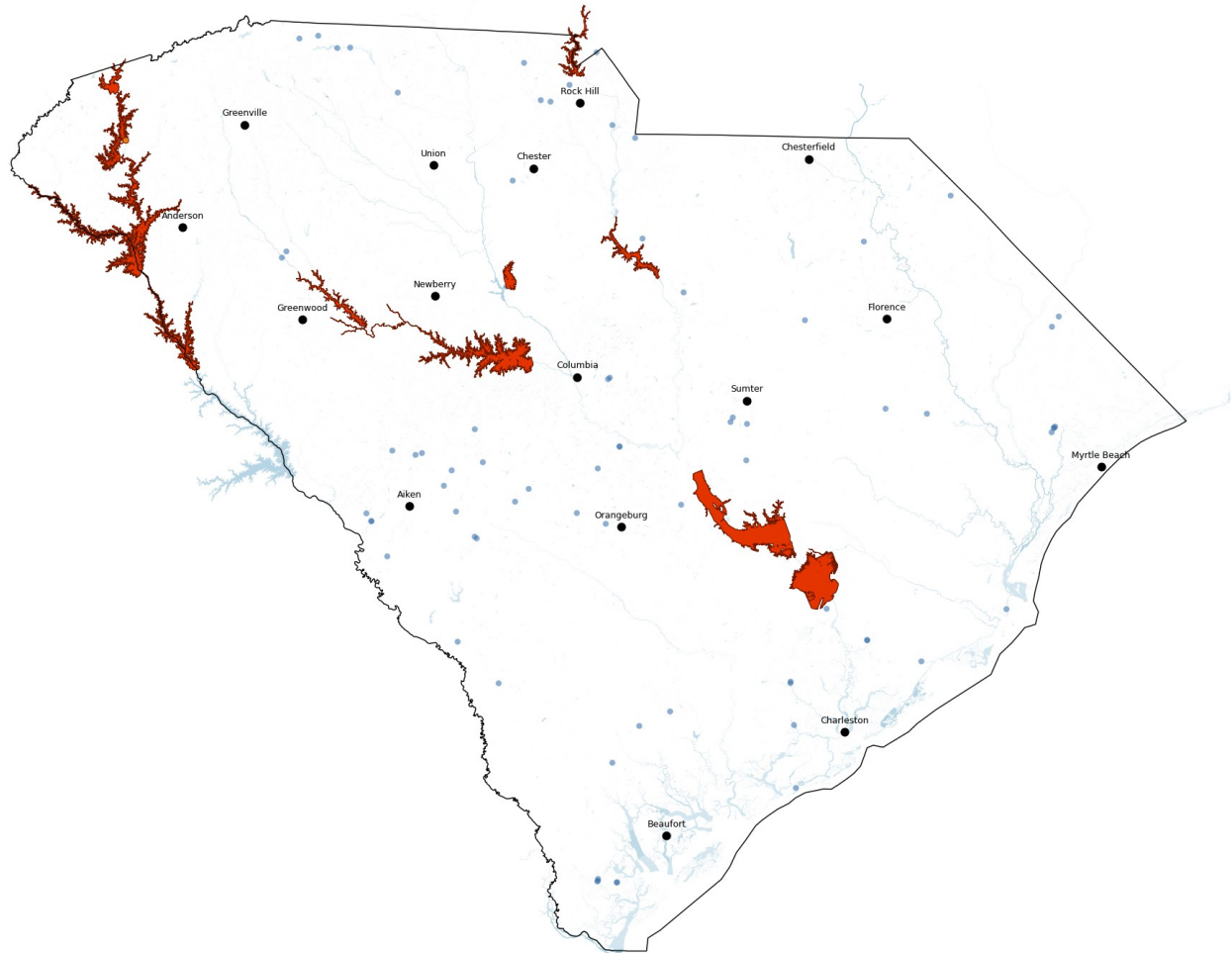


Figure 13: Planned Lakes for Data Collection

Part 3.3 Multi-Gain Data Fusion Technique

Implement the scheme of generating multi-gain radar input frames. We will produce, for example, three intensity images: one with low gain (conservative threshold), one medium, and one high gain (sensitive). These three images form a multi-channel radar representation of that moment. This step may involve developing a script or using the radar's SDK to adjust gain or threshold levels. We will validate that this approach indeed highlights targets across the frames, e.g. a small kayak might only appear in the high-gain image, whereas land or large vessels appear even in low-gain. The combination effectively spans the dynamic range of returns.

Part 3.4 CNN Model Development

Design and train a baseline convolutional neural network that ingests the multi-gain radar data (and potentially temporal sequences) to detect objects. As a starting point, we plan to implement a 1 D convolutional detector that scans each radar range-azimuth slice (similar to a Conv1D over the radar's polar coordinate image). This simple model will be trained to distinguish target returns from clutter using the pilot data. We will compare its performance to the classical 1D CA-CFAR detector as a baseline. The hypothesis is that even a basic learning-based model can achieve better sensitivity than CFAR for low-RCS targets, when provided with multi-gain inputs. As we iterate, more sophisticated models (e.g. 2 D CNN on

the Cartesian radar image or even a radar-camera fusion network) may be explored, but the initial focus is on getting a working radar-target detector.

Part 3.5 End-to-End Detection Pipeline

Integrate the radar-CNN and camera. Detected target coordinates from the radar-CNN will be mapped onto the camera frame in real-time (or offline, initially). We will implement a simple user interface to visualize results – for example, drawing radar detections as bounding boxes on the camera video. This will serve as a radar-assisted labeling tool: when we pause the video, the system can output suggested annotations for any objects the radar picked up. The pipeline will be evaluated on the pilot dataset by measuring detection rates and correctness of the suggested labels.

Part 3.6 Preliminary Evaluation and Iteration

Evaluate the performance of the radar-based detector versus traditional methods. Specifically, we will compare detection outcomes of our CNN to the radar’s standard ARPA implementation. Key metrics include probability of detection for the small test targets and false alarm rates. If the initial model underperforms in some scenarios (e.g. very rough clutter), we will analyze the failure cases and consider additional inputs or model architecture changes. We expect to demonstrate improvement over CFAR, at least in moderate conditions, as a proof-of-concept result suitable for publication.

Part 3.7 Radar-Assisted Image Labeling

As an offshoot of the detection pipeline, develop a semi-automatic labeling tool for the camera imagery. This tool will use the radar detections (after filtering out obvious false alarms) to mark regions in each video frame that likely contain an object. A human annotator can then quickly confirm the presence of an object and label its bounding box or class, rather than searching from scratch. We will apply this process to our pilot dataset to generate training labels for the camera. This not only accelerates data annotation but also yields a more diverse training set of small-object images than could reasonably be labeled manually. The effectiveness of radar cueing will be quantified by measuring the reduction in manual effort (e.g. how many proposals per hour the radar provides and what fraction are true objects). This addresses the objective of an automated labeling pipeline using radar cues.

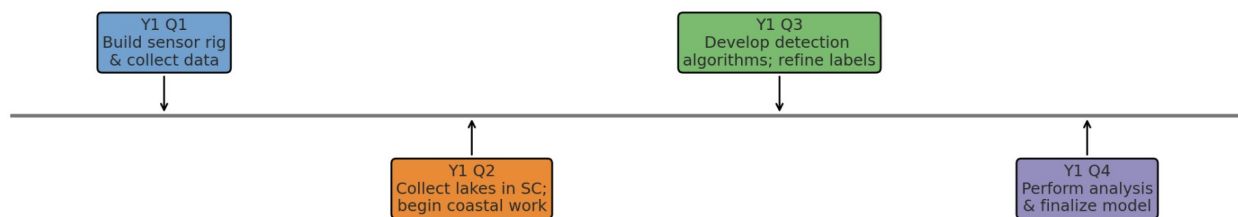


Figure 14: One-Year Timeline of Project

PART 4. EXPECTED RESULTS & OUTCOMES

Improving the detection of small objects in the maritime domain has significant real-world implications. From a maritime safety perspective, the ability to spot a person overboard or a semi-submerged hazard (e.g. a floating log or container) can save lives and prevent collisions. Small vessels such as kayaks or dinghies are often missed by large ship radars until dangerously close; a better detection system could alert operators or autonomous navigation systems earlier.

Enhancing radar detectability of these objects – without overwhelming false alarms – fills a critical gap in surveillance of ports, coastal waters, and open oceans. This is increasingly important as we move toward unmanned surface vessels and advanced maritime traffic monitoring, where reliable automation of small object detection is required for safety and security. The fusion approach in this project also offers a valuable benefit for data generation and algorithm development.

By using radar to assist in labeling optical imagery, we tackle the data scarcity issue that plagues maritime computer vision. Instead of relying on hundreds of hours of tedious manual annotation to train a camera-based detector, the radar’s cues allow us to rapidly label a high-quality dataset. This radar-assisted labeling not only reduces human workload dramatically, but it also improves dataset quality; the labels are more likely to catch hard-to-see objects that a human might overlook when fatigued. In essence, our approach leverages the complementary nature of the sensors: radar provides precision in detection cueing, and the camera provides rich detail for classification. Together, they yield a system more robust than either alone, particularly for the challenging maritime environment.

PART 5. FUTURE DIRECTION

This first phase of research will establish a strong foundation for a longer-term Ph.D. roadmap. In the future, we plan to build on the prototype and concepts developed during the Master’s work to tackle deeper challenges in radar perception and sensor fusion. Some future research directions include:

Part 5.1 Self-Supervised Radar Representation Learning

I will explore training the radar detection models with self-supervised learning techniques to reduce reliance on extensive labeled data. One planned approach is a Radar Masked Autoencoder (Radar-MAE), where the model learns to reconstruct missing portions of radar scans, thereby learning robust features from raw data. By pre-training on a large amount of unlabeled radar video (tens or hundreds of hours), the model can develop an understanding of sea clutter characteristics and typical target signatures. These features can then be fine-tuned for detection tasks with relatively few labeled examples (few-shot learning). This thrust is expected to lead to a conference publication (e.g. CVPR ’27 or ’28) showcasing that a self-supervised radar model outperforms a fully supervised one given limited labels.

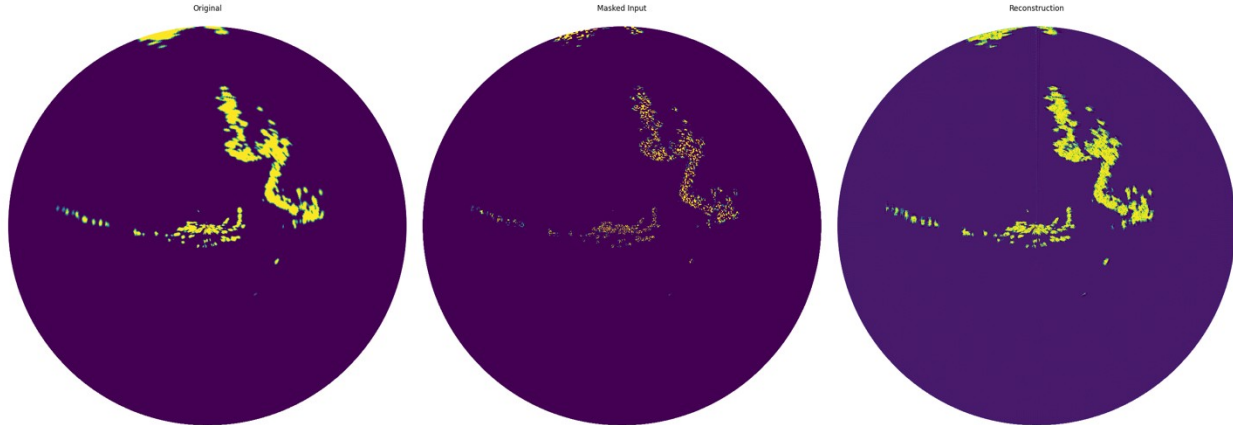


Figure 15: Results of preliminary MAE training (<10 minutes on MacBook Air)

Part 5.2 Physics-Informed and Continual Learning

Another future direction is to incorporate radar signal processing physics into the neural network architecture. For example, embedding the radar’s known range equation or CFAR principles as differentiable components (loss terms or network layers) could guide the model to make physically plausible detections (controlling false alarm rates explicitly). We will also investigate Vision Transformer (ViT) models tailored to radar by encoding priors like the pulse repetition frequency and antenna patterns. In tandem, as we deploy the system in varied environments, continual learning strategies will be needed so the model can adapt to new sea states or sensor configurations without forgetting old knowledge. Techniques like online domain adaptation and Elastic Weight Consolidation (EWC) are planned to ensure the radar perception system remains robust over time. These advanced topics form the core of much later work, aiming for journal-level contributions in the radar/remote sensing field.

Part 5.3 A-MARV Dataset Expansion

Throughout the entire project, we expect to significantly scale up our data collection to build the Amplitude-only Maritime Radar Video (A-MARV) dataset; the first open benchmark dataset of its kind. The full A-MARV will comprise many hours of real radar recordings (X-band, ~9.4 GHz) captured in diverse conditions, using a tri-gain recording strategy (temporal low/medium/high gain frames) to preserve small-target visibility. It is planned to label a portion of this data with “strong” annotations (definite object labels) and an even larger portion with “weak” annotations generated by semi-automatic means (e.g. the radar-assisted labeling plus a data programming approach using tools like Snorkel). The A-MARV dataset will be published with standardized tasks (detection, domain adaptation, continual learning, few-shot learning) to catalyze research in maritime radar ML. The goal is to eventually release A-MARV along with baseline models and an evaluation toolkit, establishing a community resource. This will not only validate the broader impact of our work but also ensure reproducibility and benchmarking for future researchers. A dedicated data descriptor paper (e.g. in IEEE T-GRS) is planned once the initial version of the dataset is ready.

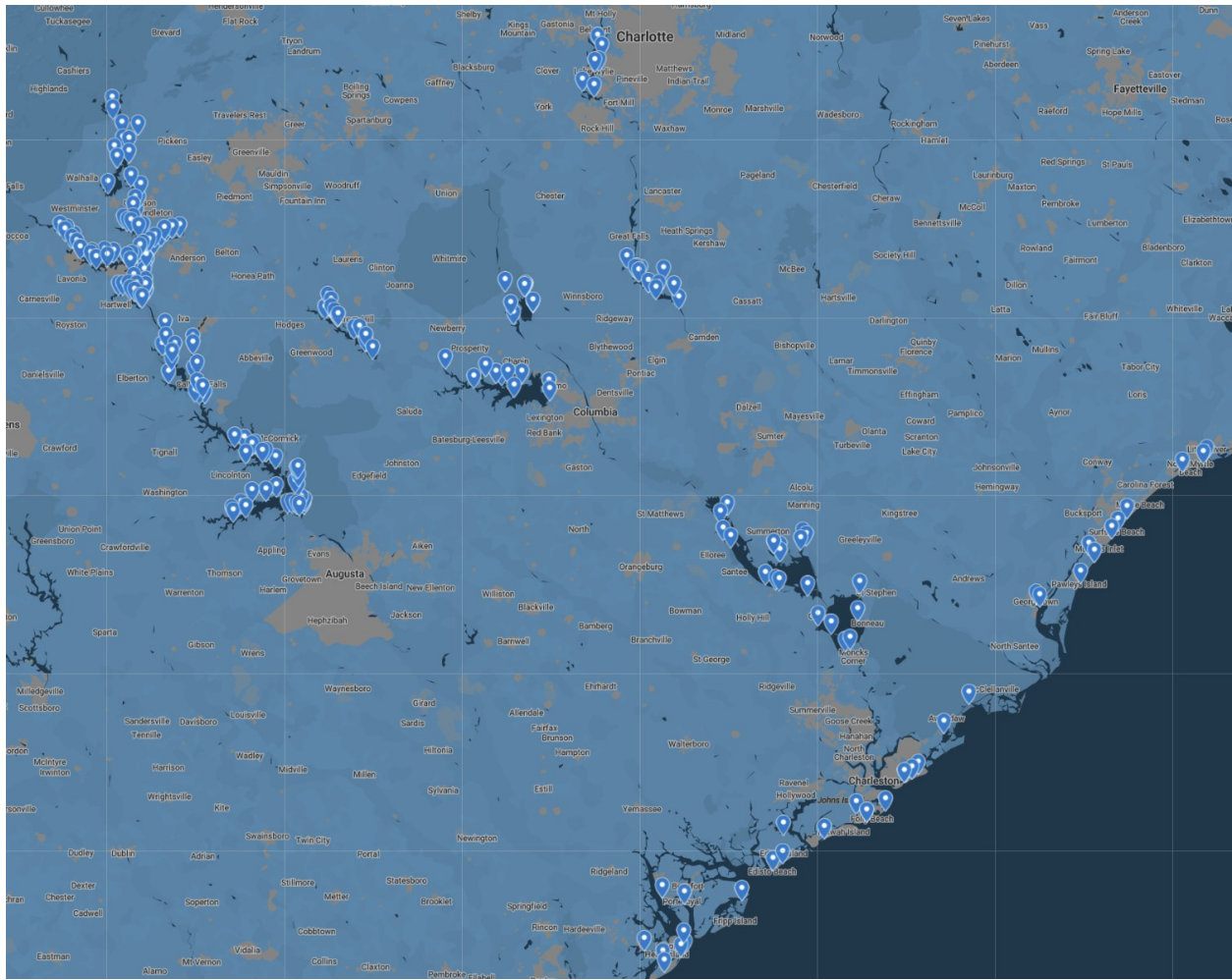


Figure 16: Initial (v0.1) A-MARV Dataset Collection Locations

So far, the core prototype has been established, integrating a multi-gain radar with high-resolution imagery and training initial detection models, and run our first small-object detection trials. Over the coming months, we'll focus on quantifying gains in small-object detection and automated labeling accuracy as well as sharing performance benchmarks and system design insights

Beyond these early milestones, we'll broaden our approach to include self-supervised learning strategies and curate a much larger, diverse dataset to drive continued improvements. Ultimately, by year five, our goal is to deliver a production-ready, low-RCS maritime-target detection system alongside the A-MARV open dataset; together enabling a more comprehensive all-weather, 24/7 maritime situational awareness.